

Sara Chaudhari

Software Engineer, Backend Systems & ML Infrastructure

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EDUCATION

University of California, San Diego, MS in Computer Science and Engineering	Sept 2024 – Jun 2026 GPA: 3.9/4.0
Relevant Coursework: Distributed Systems, Operating Systems, Computer Networks, Machine Learning Systems, Advanced Algorithms	
University of Mumbai, BE in Computer Engineering	Aug 2020 – Jun 2024 GPA: 4.0/4.0
• Honors in Artificial Intelligence and Machine Learning	

EXPERIENCE

Genies · Software Engineer Intern	Sept 2025 – Dec 2025
• Built a Python orchestration layer in front of several AI response services, adding request queuing and connection pooling so one slow downstream call no longer stalled the rest.	
• Added Redis-backed response caching for repeated queries, cutting redundant model calls by roughly 25% during peak concurrent traffic without changing response quality.	
• Wrote a shared retry-and-timeout utility that three other backend teams adopted, replacing each team’s one-off error handling with a single tested, consistent approach.	
Dassault Systèmes Americas Corp. · Software Engineer Intern	June 2025 – Sept 2025
• Built Java services managing state transitions and retries for engineering workflow jobs, containing failures so one stuck task stopped blocking the 2,000+ dependent tasks behind it.	
• Migrated a manual rule-processing step to a containerized Java job on Kubernetes, cutting average batch processing time for workflow artifacts by about 20%.	
• Wrote Jenkins-based regression tests in Groovy for the workflow engine’s core paths, catching two recurring classes of state-transition bugs before they reached release.	
UpSolve Solutions · Software Engineer	Apr 2023 – Jul 2024
• Wrote a C++ service on Linux that scored incoming user events against a pre-trained model in real time, cutting median response latency by about 30%.	
• Built a producer-consumer pipeline on Redis-backed queues to absorb traffic bursts, keeping the scoring service stable across 5,000+ requests per day at peak load.	
• Added a daily job comparing live model outputs against offline batch results, surfacing drift early and cutting time spent chasing silent scoring bugs.	
Garg Group · Software Engineer	Sept 2022 – Apr 2023
• Built Python ETL pipelines that ingested daily supply-chain files, validated schema consistency, and loaded cleaned records into a shared warehouse table for reporting.	
• Engineered rolling-window features from 100K+ inventory records to feed an anomaly-detection model that flagged irregular stock movements for the operations team.	
• Set up Airflow DAGs to run nightly scoring and refresh dashboards automatically, cutting report turnaround from about a day to under an hour.	

RESEARCH EXPERIENCE

University of California, San Diego · Graduate Researcher	Mar 2025 – Sept 2025
• Built a Kubernetes-based experimentation framework that ran 40+ training and evaluation configurations in parallel over a 2M-record dataset, cutting per-experiment setup time from hours to minutes.	
• Containerized the lab’s evaluation pipeline with Docker so results were reproducible across three lab machines, replacing a manual setup that had produced inconsistent runs.	
• Built a benchmarking harness comparing model variants on structural-recovery accuracy, showing an 18% improvement for the group’s new method over its prior baseline.	

KEY PROJECTS

Hybrid Semantic Retrieval and Ranking System	Code
• Built a hybrid search system combining BM25 with FAISS-indexed sentence embeddings, fused by reciprocal rank instead of blending incomparable raw scores.	
• Measured a 4.9% NDCG@10 and 8.1% Recall@50 gain over BM25 alone on a labeled benchmark, and load-tested indexing and query latency at 1M documents.	
Distributed Log Indexing and Search System	Code
• Built a sharded log indexing engine in Java with a parallel per-shard build phase and concurrent cross-shard query fan-out.	
• Measured a 2.83x indexing speedup over single-threaded on 6M log lines (587MB), with cross-shard query latency under 420ms.	

PUBLICATIONS

• Genome Sequence Analysis and Drug – Target Interaction Prediction using Deep Learning (Link) Springer Innovative Computing and Communications, LNNS Series (Vol. 1038)	
• Two-Pass Pipeline for Automated Medical Infographic Generation Using Text Summarization (Poster) Poster presentation, SmartDataCom: International Conference on Smart Data Processing, Communication and Networking (2024)	

SKILLS

Languages: Python, Java, C++, TypeScript, SQL	
Backend & Distributed Systems: Multithreading, Async I/O, Message Queues, Microservices, REST APIs, Linux	
Cloud & Infrastructure: AWS, Docker, Kubernetes, Terraform, Jenkins CI/CD, Airflow	
Data & ML Systems: PostgreSQL, Redis, Spark, PyTorch, Feature Pipelines, Model Serving	
Practices: Monitoring & Alerting, Automated Testing, Performance Profiling, Git	